MSK OSCE Preparation Guide

Part 1 — Cases A to D | BSc Level 5 Physiotherapy

This guide provides detailed, speakable answers for four MSK OSCE cases. Each case is answered across four examiner questions aligned with the Level 5 assessment rubric. Aim to speak clearly, use correct anatomical and clinical terminology, and always demonstrate **clinical reasoning** rather than simply listing facts. Target: Good tier (64%) with content sufficient for Very Good (70–79%).

Question	Focus Area	Key Examiner Expectation
Q1	Focused Subjective Assessment	Red flags screened; prioritised problem list using ICF/biopsychosocial model
Q2	Diagnosis & Clinical Reasoning	Primary + 2 differentials; evidence-based reasoning from signs/symptoms
Q3	Objective Assessment	2–3 techniques named, performed, interpreted correctly
Q4	Treatment & Management Plan	Initial + long-term; SMART goals; evidence-based; safety considerations

CASE A — Mrs. Gupta

50-year-old female | Graphic designer | 1-year neck pain | Osteoporosis | Spurling's & Hoffmann's negative

Q1: Focused Subjective Assessment, Red Flags & Prioritised Problem List

Opening Statement (speak this aloud):

> *"Mrs. Gupta, I'd like to start by asking you some questions about your neck pain so I can understand how it affects you day-to-day, identify anything that requires urgent attention, and then set priorities for today's assessment and treatment."*

Subjective Assessment — Key Areas to Cover

- **Chief Complaint & Body Chart:** Dull, aching neck pain — ask the patient to point to the area. Determine whether pain is local (cervical), referred (into trapezius, interscapular region), or radiating distally into the arm (dermatomal pattern suggesting radiculopathy). Mrs. Gupta is a graphic designer — prolonged static postures at a workstation are highly relevant.
- **Onset & Duration:** 1-year history — likely a chronic/recurrent mechanical condition. Ask about precipitating events: new desk setup, increased workload, ergonomic changes. Gradual onset supports postural/degenerative aetiology.
- **Pain Characteristics:** Dull quality suggests nociceptive mechanical pain. Rate severity on NRS (0–10). Assess irritability: how easily provoked, how long to settle. Low irritability allows fuller objective examination; high irritability requires a modified approach.
- **24-Hour Behaviour:** Morning stiffness < 30 minutes suggests mechanical rather than inflammatory. Aggravated by prolonged screen use, forward-head posture, overhead activity. Eased by heat, rest, movement — typical of mechanical neck pain. Night pain unrelated to position warrants red flag screening.
- **Aggravating / Easing Factors:** Cervical extension or rotation aggravated → facet or foraminal involvement. Sustained flexion aggravated → disc / deep cervical flexor fatigue. Eased by position change → mechanical; constant regardless of position → red flag.
- **Neurological Symptoms:** Ask specifically about paraesthesia, numbness, weakness in the upper limbs — even though Spurling's is negative, radiculopathy cannot be excluded by history alone. Bladder/bowel changes would indicate cervical cord involvement.
- **Psychosocial Screening (Yellow Flags):** Graphic design work may involve work pressure, repetitive strain concerns, fear-avoidance beliefs. Use STarT MSK or similar tool. Beliefs about pain ("my neck is damaged") can perpetuate disability.
- **Past Medical History & Medications:** Osteoporosis is confirmed — clarify DEXA T-score, fragility fracture history, current bone protection medication (bisphosphonates, calcium/vitamin

D). NSAIDs or analgesics currently used? Corticosteroid history (increases osteoporosis risk further).

- **Social / Functional Impact:** Driving, computer use, sleep posture, childcare/home roles. ICF: Activity limitations (turning head to reverse car) and Participation restrictions (reduced work productivity).

Red Flag Screening

The following must be screened given age, osteoporosis, and chronicity:

> ⚑ **Osteoporotic vertebral fracture:** Sudden worsening of pain, recent minor trauma, or inability to bear weight — refer immediately for imaging.

> ⚑ **Malignancy / metastatic disease:** Unexplained weight loss (>5% body weight over 3 months), night sweats, fever, constant unremitting pain unaffected by position, previous cancer history.

> ⚑ **Cervical Arterial Dysfunction (CAD):** 5Ds and 3Ns — Diplopia, Dizziness/vertigo, Dysarthria, Dysphagia, Drop attacks; Numbness (face/limbs), Nausea, Nystagmus. Screen BEFORE any cervical manual therapy.

> ⚑ **Cervical Myelopathy:** Hoffmann's is negative, but screen for hand clumsiness, wide-based gait, bilateral upper limb symptoms, hyper-reflexia.

> ⚑ **Inflammatory arthritis:** Morning stiffness > 1 hour, bilateral joint involvement, systemic features (fatigue, skin changes). Rheumatoid arthritis can cause atlantoaxial instability — a contraindication to upper cervical manipulation.

Prioritised Problem List (ICF Framework)

- **Priority 1 — Pain:** Chronic dull cervical pain NRS 5–6/10, aggravated by sustained posture. Primary driver of presentation.
- **Priority 2 — Postural Dysfunction:** Forward head posture and rounded shoulders identified on observation — contributing biomechanical factor.
- **Priority 3 — Likely Muscle Imbalance:** Deep cervical flexor (DCF) weakness + upper trapezius/levator scapulae tightness (Janda upper crossed syndrome pattern).
- **Priority 4 — Reduced Cervical ROM:** Limiting work tasks (checking blind spots, overhead filing).
- **Priority 5 — Osteoporosis:** Modifies technique selection — no high-velocity thrust to cervical/upper thoracic spine.
- **Priority 6 — Occupational Impact:** Reduced productivity and discomfort during 8-hour workday.

Q2: Primary Diagnosis, Two Differentials & Clinical Reasoning

Primary Diagnosis

Cervicogenic Neck Pain secondary to Upper Crossed Syndrome / Postural Cervical Dysfunction — mechanical nociceptive pain arising from sustained abnormal loading of cervical joints, myofascial structures, and intervertebral discs.

Clinical Reasoning:

- 1-year history consistent with a chronic mechanical condition, not an acute red-flag presentation.
- Dull, aching quality — characteristic of nociceptive mechanical pain (myofascial or articular), not neuropathic (burning/shooting) or nociplastic (widespread, disproportionate).
- Protracted head posture and rounded shoulders increase cervical axial load by up to 27 kg per inch of forward head displacement (Hansraj 2014) — directly mechanically plausible.
- Occupation as a graphic designer involves prolonged static cervical flexion/rotation — a well-established risk factor for chronic neck pain.
- Negative Spurling's (no radiculopathy) and negative Hoffmann's (no myelopathy) reduce likelihood of neurological origin.
- Osteoporosis does not cause neck pain directly but must be considered in management approach.

Differential 1: Cervical Radiculopathy (C5/C6 or C6/C7)

Radiculopathy involves nerve root compression producing dermatomal pain, paraesthesia, and myotomal weakness. Although Spurling's is negative (sensitivity ~30%, specificity ~93%), it does not fully exclude radiculopathy. Supporting features would include: unilateral arm pain in a dermatomal pattern, reduced upper limb reflexes (biceps C5/C6; triceps C7), and positive neurodynamic testing (Upper Limb Tension Test). Currently less likely given absence of distal symptoms, but must be formally cleared with objective neurological screen.

Differential 2: Inflammatory Arthritis (e.g., Rheumatoid Arthritis)

Rheumatoid arthritis (RA) affects the cervical spine in 30–40% of cases, particularly the atlantoaxial joint (C1–C2), posing a risk of cord compression. Features that would support RA: morning stiffness > 1 hour, bilateral joint swelling, extra-articular features (nodules, skin, eye involvement), elevated ESR/CRP and positive rheumatoid factor. Currently less likely given the mechanical, unilateral, posture-dependent nature of pain — but inflammatory markers and rheumatological history should be clarified, particularly before any cervical manual therapy.

Q3: Objective Assessment Techniques & Interpretation

Technique 1: Cervical Active Range of Motion (AROM) with Overpressure

- **Rationale:** Quantifies functional restriction and reproduces the patient's familiar pain — essential for baseline and outcome measurement.

- **Procedure:** Assess all 6 cervical movements: flexion, extension, left/right lateral flexion, left/right rotation. Use a goniometer or inclinometer for accuracy. Apply gentle overpressure at end range if AROM is full and pain-free.
- **Normal Values:** Flexion 45–50°, extension 55–70°, lateral flexion 45°, rotation 60–80°.
- **Interpretation:** Restriction in extension and ipsilateral rotation suggests facet joint involvement. Restriction in flexion with end-range pain suggests discogenic or deep cervical flexor involvement. Deviation on rotation suggests muscle guarding or unilateral facet hypomobility.

Technique 2: Deep Cervical Flexor (DCF) Endurance Test — Craniocervical Flexion Test (CCFT)

- **Rationale:** DCF weakness (longus colli, longus capitis) is the hallmark of cervicogenic neck pain and upper crossed syndrome. These muscles stabilise the upper cervical spine and correct forward head posture.
- **Procedure:** Patient supine. Inflate pressure biofeedback unit (Chattanooga stabiliser) to 20 mmHg baseline under cervical lordosis. Instruct patient to perform gentle nodding action (craniocervical flexion — NOT gross neck flexion). Progressively target 22, 24, 26, 28, 30 mmHg in 2 mmHg increments, holding each for 10 seconds.
- **Interpretation:** Inability to reach 26 mmHg without compensatory superficial flexor activity (sternocleidomastoid, anterior scalene) indicates DCF weakness — common in chronic neck pain. Normal is achieving 28–30 mmHg with smooth activation.

Technique 3: Upper Trapezius / Levator Scapulae Length Test

- **Rationale:** In Janda's Upper Crossed Syndrome, upper trapezius and levator scapulae are hypertonic (overactive/shortened), contributing to elevated and protracted shoulders and increased cervical loading.
- **Procedure:** Stabilise shoulder girdle (depress and laterally rotate scapula). Passively laterally flex the cervical spine away from the side being tested. Measure range and assess for stretch pain, tissue resistance, or asymmetry.
- **Interpretation:** Reduced ipsilateral lateral flexion with reproduction of stretch pain/resistance confirms tightness in upper trapezius/levator scapulae. Combined with DCF weakness, confirms the upper crossed syndrome pattern — informing the priority to stretch and inhibit these muscles while strengthening DCF and lower trapezius.

Q4: Initial & Long-Term Treatment / Management Plan

SMART Goals

- **Short-term (4 weeks):** Reduce neck pain from NRS 6/10 to ≤ 3/10 at work; improve cervical rotation ROM to within 10° of normal.
- **Long-term (12 weeks):** Patient independently manages posture at workstation; DCF endurance to 28 mmHg on CCFT; return to full work productivity without pain.

Initial Management (Sessions 1–3)

- **Patient Education:** Explain the postural mechanism — forward head posture increases cervical load and compresses facet joints. Dispel catastrophising (pain does not equal damage). Introduce self-management concepts: movement breaks every 30–45 minutes, postural cues.
- **Pain Management:** Heat/TENS for symptomatic relief. Heat over upper trapezius before exercise to reduce muscle tone. TENS at conventional (80–100 Hz) for gate control analgesia.
- **Manual Therapy — Modified for Osteoporosis:** Grade II–III posteroanterior (PA) cervical mobilisation (Maitland) to restricted segments — gentle oscillatory technique appropriate with osteoporosis. NO high-velocity thrust manipulation to cervical or upper thoracic spine. Soft tissue massage / trigger point therapy to upper trapezius and levator scapulae to reduce muscle hypertonicity.
- **Ergonomic Advice:** Screen at eye level, monitor at arm's length, lumbar support, forearm support to reduce scapular elevation, frequent movement breaks.

Progressive Rehabilitation (Sessions 4–12)

- **Therapeutic Exercise — DCF Strengthening:** CCFT-based progressive programme — start at 22 mmHg, progress by 2 mmHg per week as tolerated. Sets/reps: 3 × 10 holds at each pressure level. Progress to functional patterns (supine → sitting → standing).
- **Scapular Stabilisation:** Lower trapezius strengthening (prone Y, W exercises), serratus anterior activation — to correct rounded shoulders and reduce upper trapezius overload.
- **Cervical Flexibility:** Upper trapezius/levator scapulae stretching (30-second holds, 3 reps, daily) — Muscle Energy Technique (MET) for active participation. Cervical retraction exercises (chin tuck) to correct forward head posture.
- **Postural Taping:** Kinesiotape or postural correction tape to facilitate scapular retraction and reduce forward head posture during work — short-term adjunct while motor learning is established.
- **Bone Health Liaison:** Confirm liaison with GP regarding osteoporosis management (DEXA monitoring, medication review, fall prevention strategies). Bone-loading exercises (upright weight-bearing postures) are beneficial for bone density.
- **Discharge Planning:** Home exercise programme (HEP) reinforced from session 1. App or written record for exercise adherence. Flare-up management plan. GP referral if symptoms worsen or neurological signs develop.

> *⚠ Safety Note: Given osteoporosis, avoid cervical axial loading, high-velocity thrust techniques, and sustained end-range compression. All manual therapy must be graded gently with ongoing reassessment.*

CASE B — Mr. O'Shea

72-year-old male | Retired librarian | 10-month thoracic back pain | Osteoporosis | Corticosteroids | Percussion pain positive at T7–T8

Q1: Focused Subjective Assessment, Red Flags & Prioritised Problem List

Opening Statement:

> *"Mr. O'Shea, before I examine your back, I'd like to ask you some important questions. Some of your details — particularly your age, your bone condition, and your medication — mean I need to be especially thorough about understanding your symptoms and ruling out anything that needs urgent attention."*

Subjective Assessment — Key Areas

- **Chief Complaint & Location:** Thoracic back pain centred at T7–T8 (mid-thoracic). Determine whether pain is localised (one segment) or diffuse. Ask about radiation: intercostal pattern (nerve root), anterior chest (referred visceral), or down into the lower back/legs.
- **Onset:** 10 months — gradual or sudden? A sudden onset after minimal trauma (bending, lifting light object, coughing, sneezing) in a patient with osteoporosis is HIGHLY suspicious for a vertebral compression fracture (VCF).
- **Increasing Kyphosis:** A change in posture over the period of symptoms suggests progressive vertebral collapse — hallmark of VCF. Ask if height has reduced.
- **Pain Quality:** Sharp, knife-like thoracic pain that is worse on movement and weight-bearing is consistent with a fracture. Dull, aching pain suggests muscular or degenerative aetiology. Night pain that wakes from sleep or constant unremitting pain → red flag.
- **Aggravating / Easing Factors:** VCF characteristically worsens with spinal loading (standing, walking, transitional movements — lying to sitting) and eases with recumbency. Degenerative disc pain is often eased by walking (extension-biased).
- **Corticosteroid History:** Duration, dose, reason for prescription (COPD? Inflammatory condition?). Long-term corticosteroids (> 3 months, > 5mg prednisolone/day) cause significant bone loss — markedly increasing fracture risk.
- **Neurological Symptoms:** Any lower limb weakness, numbness, bilateral leg symptoms, gait disturbance, or bowel/bladder changes → EMERGENCY. A VCF can result in spinal cord compromise (thoracic cord compression → paraparesis).
- **Systemic Symptoms:** Unexplained weight loss, fatigue, fever, night sweats — screen for malignancy (multiple myeloma, metastatic disease very common in 70+ male with thoracic pain).

- **Previous Cancer History:** Thoracic spine is one of the most common sites for skeletal metastases. Prostate, lung, kidney, thyroid, and breast cancers commonly metastasise to the spine.
- **Functional Impact:** Increasing kyphosis limits standing tolerance, may impair respiratory function (thoracic cage restriction), affects dignity and independence with ADLs.

Red Flag Screening — HIGH PRIORITY CASE

> **This patient has multiple concurrent red flags. Immediate screening is essential before any assessment or treatment proceeds.**

> 🚩 **VERTEBRAL COMPRESSION FRACTURE (Primary concern):** Osteoporosis + corticosteroids + percussion pain at T7–T8 + increasing kyphosis = highly suspicious. Percussion test positive at T7–T8 is a direct indicator.

> 🚩 **MALIGNANCY (Metastatic spinal disease):** Age 72, male. Thoracic pain with systemic features (weight loss, night sweats) must be assumed malignant until proven otherwise. Multiple myeloma and prostate metastases are common in this demographic.

> 🚩 **SPINAL CORD COMPRESSION:** Any lower limb weakness, bilateral sensory change, or bladder/bowel dysfunction = neurological emergency. Do NOT continue assessment — refer immediately to ED.

> 🚩 **CORTICOSTEROID-INDUCED OSTEOPOROSIS:** Significantly elevates fracture risk above baseline osteoporosis.

Prioritised Problem List

- **Priority 1 — URGENT: Rule out/confirm VCF and malignancy:** Percussion pain at T7–T8 + clinical picture = immediate referral for imaging (plain X-ray minimum; MRI if malignancy suspected or neurological signs present).
- **Priority 2 — Pain:** Thoracic pain limiting all functional activities.
- **Priority 3 — Postural deterioration:** Increasing kyphosis risking respiratory compromise and further falls.
- **Priority 4 — Falls Risk:** Increased kyphosis shifts centre of gravity anteriorly, increasing falls risk in an elderly patient with osteoporosis.
- **Priority 5 — Functional decline:** Reduced independence with ADLs, mobility, and community participation.

Q2: Primary Diagnosis, Two Differentials & Clinical Reasoning

Primary Diagnosis

Osteoporotic Vertebral Compression Fracture (VCF) at T7–T8 — fragility fracture of one or more thoracic vertebral bodies due to compromised bone mineral density, resulting in anterior vertebral

collapse, increasing kyphosis, and localised percussion pain.

Clinical Reasoning:

- Percussion pain directly over T7–T8 spinous processes is a sensitive indicator of vertebral fracture — it transmits a compressive force through the vertebral body rather than the posterior elements.
- Osteoporosis + long-term corticosteroids = high fracture risk (corticosteroids suppress osteoblast activity and enhance osteoclast activity).
- Progressive kyphosis over 10 months is consistent with anterior vertebral body collapse — 'wedge fracture' appearance on imaging.
- Male sex, age 72 — men account for approximately 30% of VCFs, often under-diagnosed due to lower screening rates than women.
- Retired librarian — relatively sedentary; reduced weight-bearing mechanical stimulus reduces bone density maintenance.

Differential 1: Malignant Spinal Disease (Metastatic / Multiple Myeloma)

Skeletal metastases to the thoracic spine are common in men over 65, particularly from prostate, lung, or renal primary cancers. Features distinguishing metastatic disease from osteoporotic VCF include: pain at rest or night pain, systemic features (weight loss, fatigue), multiple levels affected on imaging, bone scan hot-spots, elevated ESR/calcium/alkaline phosphatase. This differential CANNOT be excluded clinically and requires urgent imaging. Multiple myeloma should also be considered — it causes lytic (punched-out) lesions on X-ray and presents with bone pain, anaemia, and elevated paraprotein.

Differential 2: Degenerative Thoracic Disc Disease / Facet Osteoarthritis

Degenerative changes are universal in this age group and may produce thoracic stiffness and diffuse aching. However, degenerative disease alone does not explain: (a) percussion pain localised to one segment, (b) progressive kyphotic deformity, or (c) the severity and 10-month progressive course. Degenerative changes are likely co-existent but are a secondary rather than primary diagnosis. This distinction is critical — degenerative disease may be managed with physiotherapy while a fracture or malignancy requires a different pathway.

Q3: Objective Assessment Techniques & Interpretation

> ⚠ *Important: Objective assessment must be MODIFIED and CAUTIOUS given the high likelihood of VCF. No spinal loading tests (e.g., compression tests), no manipulative procedures. Prioritise safe screening techniques.*

Technique 1: Spinal Percussion Test

- **Rationale:** Directly assesses bone integrity of the vertebral body — the primary clinical test for VCF.

- **Procedure:** Patient seated or prone (prone preferred if tolerated). Use fist or reflex hammer to apply gentle direct percussion over each thoracic spinous process from T1 downward. Assess for localised pain reproduction versus generalised tenderness.
- **Interpretation:** Localised sharp pain at T7–T8 on percussion = high suspicion for fracture or malignancy at that level. Compare with adjacent levels — isolated response supports structural integrity failure at that segment. This finding alone warrants urgent imaging referral.

Technique 2: Thoracic Active Range of Motion — Observational / Pain-Free Only

- **Rationale:** To quantify movement restriction and identify directional pain provocation — guiding rehabilitation once fracture is confirmed/excluded.
- **Procedure:** Observe active thoracic flexion, extension, bilateral rotation, and lateral flexion. DO NOT apply overpressure given fracture suspicion. Note quality of movement, spinal rhythm, presence of a fixed kyphotic segment.
- **Interpretation:** Reduced thoracic extension (commonly < 20°) with pain at T7–T8 during extension is consistent with VCF — anterior vertebral fractures are aggravated by extension loading. A fixed kyphotic segment suggests established vertebral collapse. Global restriction in all planes may indicate co-existing degenerative disease.

Technique 3: Lower Limb Neurological Screen

- **Rationale:** T7–T8 is at the level of the spinal cord. A fracture or malignant lesion at this level can compress the cord, causing upper motor neurone signs below the level of injury.
- **Procedure:** Lower limb myotomes (hip flexion L1/L2, knee extension L3/L4, ankle dorsiflexion L4/L5, plantarflexion S1/S2). Lower limb deep tendon reflexes (knee jerk L3/L4, ankle jerk S1). Sensation in lower limb dermatomes (light touch, pinprick). Babinski's sign.
- **Interpretation:** Any weakness, reflex change, sensory loss, or positive Babinski's indicates cord compromise — this is a NEUROLOGICAL EMERGENCY requiring immediate ED referral. Normal findings allow continued physiotherapy assessment pending imaging results.

Q4: Initial & Long-Term Treatment / Management Plan

> **DO NOT initiate manual therapy or spinal exercise until imaging confirms the diagnosis. The clinical picture (percussion pain + osteoporosis + corticosteroids + progressive kyphosis) constitutes a medical red flag requiring urgent investigation.**

Immediate Management — Referral Pathway

- **Urgent Referral:** Write to GP or refer to Emergency Department (if neurological signs present) requesting: AP and lateral thoracic spine X-ray minimum; MRI thoracic spine if malignancy or cord compromise suspected.
- **Safety Netting:** Provide clear safety-netting advice — return immediately to ED if: new leg weakness, loss of bladder/bowel control, inability to stand, fever.

If VCF Confirmed (Non-Surgical, Stable Fracture)

- **Pain Management (Short Term):** Paracetamol (preferred in elderly — NSAID risk of GI bleed, renal impairment). Heat therapy for paraspinal muscle spasm. TENS over thoracic paraspinal muscles. Brief relative rest (not bed rest) for 1–2 weeks.
- **ABSOLUTE CONTRAINDICATIONS:** High-velocity thrust manipulation to thoracic spine; axial loading exercises; extension-based exercises initially; Grade IV–V mobilisation; traction.
- **Early Mobilisation (within 1–2 weeks):** Gentle supported walking to prevent deconditioning. Supine and seated pain-free movements. Bed transfer and positional advice to reduce pain during transitional movements.

Long-Term Rehabilitation (Once Fracture Stabilises — 6–12 Weeks)

- **Bone Health Optimisation:** Liaise with GP/rheumatologist for bisphosphonate prescription, calcium (1200 mg/day), vitamin D (800–1000 IU/day). Consider referral for vertebroplasty/kyphoplasty if pain is refractory.
- **Postural Rehabilitation:** Gentle thoracic extension exercises (prone on elbows progression) to reduce kyphotic progression. Avoid flexion-dominant exercises — flexion increases anterior vertebral body compression. Wall slides for scapular retraction.
- **Strength Training:** Low-load paraspinal extensor strengthening (bird-dog, prone hip extension). Avoid high spinal loading. Hip and knee strengthening to reduce falls risk.
- **Falls Prevention:** Balance training (static → dynamic). Home hazard assessment. Referral to occupational therapy for aids and adaptations. Footwear review.
- **Respiratory Physiotherapy:** Increasing kyphosis compromises thoracic cage expansion — breathing exercises, incentive spirometry if required.
- **SMART Goals:** At 12 weeks: walking 20 minutes without rest; pain ≤ 3/10 NRS; independent home exercise programme; no further falls.

> *Key clinical message: This case is primarily a MEDICAL RED FLAG scenario. Demonstrating thorough red flag reasoning and appropriate referral decision-making is the highest-priority clinical skill being assessed here.*

CASE C — Ms. Taylor

32-year-old female | Yoga instructor | 6 months postpartum | Right SIJ / buttock pain | Thigh Thrust & Sacral Thrust positive | SLR & FADDIR negative

Q1: Focused Subjective Assessment, Red Flags & Prioritised Problem List

Opening Statement:

> *"Ms. Taylor, I understand you've been having right-sided low back and buttock pain since having your baby six months ago. I'd like to ask you some detailed questions before examining you — including some questions to make sure we're not missing anything important. I'll also want to understand how this is affecting your work as a yoga instructor and your life at home."*

Subjective Assessment — Key Areas

- **Chief Complaint & Body Chart:** Right-sided LBP with buttock pain. Clarify exact location — SIJ pain typically localises to the PSIS region and gluteal area, rarely below the knee. Draw pain distribution: unilateral posterior pain with possible posterior thigh referral is characteristic of SIJ dysfunction.
- **Onset & Context:** 6 months postpartum — the pelvic girdle has undergone significant hormonal and biomechanical change during pregnancy. Relaxin (produced during pregnancy and lactation) increases ligamentous laxity across the pelvic girdle including the SIJ and symphysis pubis. This persists into the postpartum period.
- **Pain Quality:** Dull aching at rest, sharp on movement — consistent with joint/ligament-mediated nociceptive pain. Weight-bearing activities (standing, walking, stair climbing) typically aggravate SIJ dysfunction.
- **Aggravating Factors:** Rolling in bed, walking, stairs, prolonged standing, asymmetric loading (standing on one leg, hip thrust yoga postures). These all involve SIJ loading or shear. Carrying the baby on one hip is a classic aggravating factor.
- **Easing Factors:** Rest, lying symmetrically, SIJ belt. Pain that improves with an SIJ belt strongly supports SIJ aetiology.
- **24-Hour Behaviour:** Morning pain/stiffness common with pelvic girdle pain (PGP). Ask about sleep position — lying on the painful side or in unsupported postures aggravates SIJ.
- **Obstetric History:** Mode of delivery (vaginal/caesarean), birth complications, perineal trauma. Breastfeeding (prolongs relaxin elevation). Previous PGP in pregnancy?
- **Pelvic Floor:** As a postpartum patient and yoga instructor, pelvic floor dysfunction (weakness or hypertonia) is relevant. Ask about urinary incontinence, prolapse symptoms, pain with intercourse — these may co-exist and influence core stability.

- **Yoga Practice:** What postures/activities aggravate? Hip openers (pigeon pose, wide-legged forward fold) load the SIJ. Has she returned to teaching — relevant functional goal?
- **Psychosocial:** New motherhood is a significant psychosocial stressor. Sleep deprivation, postnatal mood disturbance, body image concerns. Ask about mood and support network.

Red Flag Screening

> 🚩 **Cauda Equina Syndrome:** Bilateral lower limb weakness/numbness, saddle anaesthesia, loss of bladder/bowel control — emergency. SLR is negative but cauda equina remains a red flag to clear.

> 🚩 **DVT / Vascular:** Postpartum period carries elevated DVT risk (hypercoagulable state persists 6 weeks to 3 months post-delivery). Unilateral calf swelling, redness, warmth — if present, do NOT mobilise: refer immediately.

> 🚩 **Pubic Symphysis Diastasis:** Symphysis pubis pain (anterior pelvic pain), waddling gait, pain with ASLR — requires imaging to exclude.

> 🚩 **Sacral Insufficiency Fracture:** Rare but occurs postpartum, especially if nutritional deficiency (calcium, vitamin D) during lactation. Severe constant pain not responding to rest.

> 🚩 **Postpartum Septic Arthritis / Discitis:** Fever, night sweats, non-mechanical unremitting pain — rare but possible if recent instrumental delivery or epidural.

Prioritised Problem List

- **Priority 1 — SIJ Pain and Pelvic Girdle Instability:** Right-sided buttock/LBP, positive provocation cluster, postpartum ligamentous laxity.
- **Priority 2 — Pelvic Floor Dysfunction:** Likely weakness or incoordination contributing to reduced lumbopelvic stability (inner unit: pelvic floor, TrA, multifidus, diaphragm).
- **Priority 3 — Hip Abductor / Gluteal Weakness:** Possible Trendelenburg sign — gluteus medius weakness leads to contralateral pelvic drop, increasing SIJ shear forces.
- **Priority 4 — Occupational and Functional Limitation:** Unable to return to full yoga teaching, limited with childcare tasks.
- **Priority 5 — Psychosocial:** New motherhood stressors, concern about return to professional practice.

Q2: Primary Diagnosis, Two Differentials & Clinical Reasoning

Primary Diagnosis

Postpartum Sacroiliac Joint Dysfunction (Pelvic Girdle Pain — PGP) — ligamentous hypermobility of the right SIJ mediated by relaxin, resulting in abnormal joint mechanics, pain provocation on loading tests, and functional disability.

Clinical Reasoning:

- Positive Thigh Thrust and Sacral Thrust tests — these are two of the five tests in the Active SIJ Provocation Cluster (Laslett 2005). A cluster of ≥ 3 positive tests has high specificity for SIJ pain. Two positive tests still significantly increases the probability of SIJ origin.
- Negative SLR: lumbar nerve root compression (L4/L5/S1) effectively excluded — the most common alternative cause of unilateral LBP radiating to the buttock in this age group.
- Negative FADDIR: hip intra-articular pathology (labral tear, FAI) less likely — important differential given age and yoga practice.
- Postpartum context: Relaxin levels remain elevated during lactation, maintaining ligamentous laxity beyond delivery. Asymmetric sacroiliac loading (carrying baby on one side) creates shear forces that stress the already-lax SIJ.
- Pain localised to PSIS/gluteal area — typical SIJ pain pattern. Does not radiate below the knee (distinguishing from discogenic/radicular).

Differential 1: Lumbar Facet Joint Pain / Discogenic Pain

Lumbar pathology (disc prolapse, facet arthropathy) commonly presents with unilateral LBP and gluteal referral. Distinguishing features would include: centrally located lumbar pain, pain worse with lumbar rotation/extension, positive lumbar provocation tests, and positive neurodynamic testing (SLR, femoral nerve). SLR is negative in this case, reducing likelihood of significant discogenic/radicular involvement. However, lumbar segmental dysfunction may co-exist with SIJ dysfunction, particularly L4–L5 and L5–S1 which are adjacent to the SIJ.

Differential 2: Hip Intra-Articular Pathology (Labral Tear / FAI)

As a yoga instructor performing frequent end-range hip movements, Ms. Taylor is at risk for femoroacetabular impingement (FAI) or acetabular labral tear. FADDIR is negative, making significant anterior FAI less likely — however, FADDIR has moderate sensitivity/specificity and does not exclude posterior labral pathology. Features suggesting hip pathology would include: groin pain (anterior > posterior), pain with hip internal rotation/flexion combined, clicking or locking sensation, pain with prolonged sitting. This remains a differential requiring complete hip examination.

Q3: Objective Assessment Techniques & Interpretation

Technique 1: Thigh Thrust Test (Posterior Shear Test)

- **Rationale:** One of the highest-sensitivity SIJ provocation tests. Applies a posteriorly-directed shear force through the SIJ via the femur.
- **Procedure:** Patient supine. Flex hip to 90° on the test side. Physiotherapist stabilises contralateral ASIS with one hand. Apply a posteriorly-directed force through the femur (axial compression through the hip in line with the femoral shaft). A positive test reproduces the patient's familiar posterior pelvic/buttock pain.
- **Interpretation:** Reproduction of posterior pelvic pain = positive. Sensitivity ~88%, specificity ~69% for SIJ as pain source. Must correlate with sacral thrust result (cluster approach). Pain in

the groin or anterior hip would suggest hip joint pathology rather than SIJ.

Technique 2: Active Straight Leg Raise (ASLR) Test

- **Rationale:** The ASLR assesses the ability of the lumbopelvic stability system to transfer load from the spine to the lower limb through the SIJ — a key function impaired in PGP. It is also a validated outcome measure in postpartum PGP.
- **Procedure:** Patient supine, legs straight. Ask patient to raise one leg 20 cm from the table without bending the knee. Assess for: difficulty/heaviness, provocation of posterior pelvic pain, pelvic rotation. Repeat with the physiotherapist manually compressing both iliac crests toward the midline (simulating an SIJ belt). A positive test is made significantly easier with compression.
- **Interpretation:** Difficulty raising the affected leg (rated 0–5 by patient) compared to the other side indicates impaired load transfer. Improvement with manual compression (or belt) = evidence of SIJ instability (force closure deficit). This directly informs the use of an SIJ belt in early management.

Technique 3: Gluteus Medius Strength Test (Resisted Hip Abduction)

- **Rationale:** Gluteus medius is essential for pelvic stability during single-leg stance (load transfer across the SIJ). Weakness increases SIJ shear force and is both a contributor and consequence of PGP.
- **Procedure:** Patient side-lying with test leg uppermost, hip in neutral rotation, knee extended. Physiotherapist stabilises pelvis at the iliac crest. Resist hip abduction with one hand applied just above the knee. Grade using Oxford Scale (0–5). Compare bilaterally. Observe for pelvis hiking (TFL compensation).
- **Interpretation:** Weakness (Oxford ≤ 4/5) or pain on resisted hip abduction on the right (affected side) confirms a biomechanical driver of SIJ stress. A Trendelenburg test (contralateral pelvic drop on single-leg stance) provides functional correlate of gluteus medius weakness.

Q4: Initial & Long-Term Treatment / Management Plan

SMART Goals

- **Short-term (4–6 weeks):** Reduce right buttock pain from NRS 6/10 to ≤ 3/10 during childcare tasks; able to perform 45-minute yoga session without pain exacerbation.
- **Long-term (12–16 weeks):** Full return to yoga teaching (including standing postures); pain-free SIJ provocation cluster; independent core stability programme.

Initial Management (Weeks 1–3)

- **SIJ Belt / Pelvic Girdle Support:** Prescribe a pelvic girdle belt worn at the level of the greater trochanters (NOT the ASIS — that is a lumbar support). Worn during aggravating activities. Supported by ASLR improvement with compression. Important: belt is a temporary support for symptom control, not a permanent solution.

- **Load Management and Postural Advice:** Avoid asymmetric loading: carry baby centrally (front carrier), avoid standing on one leg, step-over gait, single-leg yoga postures initially. Transition movements symmetrically (roll from lying to sitting with knees together).
- **Pelvic Floor Rehabilitation:** Refer to specialist pelvic health physiotherapist if bladder/bowel symptoms present. Begin pelvic floor activation exercises — gentle contraction of slow-twitch fibres (hold 10 seconds) and fast-twitch fibres (quick flicks). Coordinate with breathing (inhale, exhale then contract).
- **Pain Education:** Explain the hormonal mechanism (relaxin, ligamentous laxity) — helps the patient understand why pain is present and that it will improve. Reduce catastrophising. Normalise gradual return to activity.

Progressive Rehabilitation (Weeks 4–16)

- **Lumbopelvic Stability Programme:** Transversus abdominis (TrA) activation in supine, progressing to quadruped, sitting, standing. Multifidus activation (prone lying, bird-dog). Integrate pelvic floor + TrA co-activation — the inner unit.
- **Gluteus Medius Strengthening:** Side-lying hip abduction (Oxford Scale progression), clamshells (hip abduction + ER in side-lying), standing hip abduction with band resistance, single-leg bridges. Progress to functional patterns: split squats, step-ups.
- **Gluteus Maximus Strengthening:** Bridges → single-leg bridges → hip thrusts (monitor for SIJ provocation). Deadlift pattern with load once stable.
- **Return to Yoga:** Graded return — supine and seated postures first, avoid deep hip openers (pigeon, wide-legged forward fold) until gluteal strength restored. Work with the patient as a yoga professional — she will understand anatomical cuing. Avoid eka pada rajakapotasana (pigeon) until asymmetric hip loading is pain-free.
- **Manual Therapy — Adjunct:** Grade II SIJ mobilisation (gentle anterior/posterior glide) for pain relief if required. Soft tissue release to piriformis/gluteal muscles — can become hypertonic in compensation for SIJ instability. Note: Grade III–IV SIJ mobilisation may be contraindicated if ligamentous laxity is the primary driver — avoid hypermobility stress.
- **Discharge:** When ASLR negative, SIJ provocation cluster negative, and gluteus medius strength ≥ 4+/5 bilaterally. Provide written HEP. Advise that PGP may recur in subsequent pregnancy — early physiotherapy referral recommended.

> *Key clinical message: The positive provocation cluster (Thigh Thrust + Sacral Thrust) is the diagnostic cornerstone. Management focuses on force closure (active stability) rather than passive support alone. Inner unit and outer unit rehabilitation is the evidence-based approach.*

CASE D — Mr. Rossi

45-year-old male | Marathon runner | 3-month deep left buttock pain | Positive FADIR | Resisted hip abduction painful | Negative lumbar clearance

Q1: Focused Subjective Assessment, Red Flags & Prioritised Problem List

Opening Statement:

> *"Mr. Rossi, I'd like to start by getting a detailed picture of your left buttock pain — when it started, exactly where it is, what makes it better or worse, and how it's been affecting your running and daily life. I'll also ask a few important questions to make sure there's nothing we need to address urgently before we proceed."*

Subjective Assessment — Key Areas

- **Chief Complaint & Location:** Deep left buttock pain — the key word is 'deep'. This distinguishes from superficial gluteal muscle pain. Deep buttock pain suggests: piriformis/sciatic nerve interface, proximal hamstring origin (ischial tuberosity), ischial bursa, gluteus medius/minimus tendinopathy, or deep gluteal space pathology. Ask the patient to point — pain directly over the ischium suggests proximal hamstring; pain in the belly of the buttock suggests piriformis/deep gluteal space.
- **Onset & Mechanism:** 3-month history. Marathon running context — ask specifically about training load changes: Did mileage increase significantly before onset? Did he add hill training, speed work, or change footwear? Overuse injuries in runners typically follow a training load spike (chronic:acute workload ratio > 1.5).
- **Relationship to Running:** At what point in the run does pain onset? Deep gluteal syndrome (DGS) / piriformis syndrome typically worsens as run duration increases (ischaemic compression of the sciatic nerve). Pain that prevents completion of long runs is consistent. Post-run stiffness vs. constant pain at rest are important distinctions.
- **Sciatic-type Symptoms:** Deep buttock pain with posterior thigh radiation suggests sciatic nerve involvement within the deep gluteal space (non-discogenic sciatica). Ask about: tingling, burning, or numbness down the posterior thigh and leg. This distinguishes from lumbar radiculopathy — lumbar clearance is already negative.
- **Sitting Pain:** Piriformis syndrome is classically aggravated by prolonged sitting (compresses sciatic nerve against the piriformis). 'Wallet-sign' — sitting on a wallet in the back pocket. Pain with sitting that radiates down the leg is a hallmark.
- **Activities of Daily Living:** Stair climbing, hip internal rotation/adduction movements (crossing legs, getting in/out of car), prolonged sitting at work.

- **Previous Injury History:** Prior gluteal, hamstring, or hip injuries. Previous hip arthroscopy (suggests labral history). Previous lumbar disc herniation.
- **Psychosocial / Yellow Flags:** Running is likely an important part of Mr. Rossi's identity and mental health. Fear of not returning to marathon running, kinesiophobia, catastrophising — assess using psychosocial screening.

Red Flag Screening

> **Gluteal / Pelvic Malignancy:** Deep, constant, unremitting buttock pain unrelated to activity, particularly in a 45-year-old male, warrants consideration of soft tissue sarcoma, rectal/sigmoid colon malignancy, or prostate metastasis. Weight loss, fever, non-mechanical pain.

> **Intrapelvic Pathology:** Bowel/bladder symptoms with deep pelvic pain — could indicate abscess, haematoma, or nerve entrapment from a pelvic mass.

> **Femoral Neck Stress Fracture:** In a high-mileage runner, femoral neck stress fracture must be excluded. Groin pain worse with weight-bearing, positive hop test, night pain. If suspected — do NOT allow continued running; urgent X-ray/MRI.

> **Vascular Claudication:** Deep buttock pain that mimics claudication (pain with walking that resolves with rest, not with position change) — consider iliac artery endofibrosis (rare in runners) or peripheral vascular disease.

Prioritised Problem List

- **Priority 1 — Deep Gluteal Pain and Likely Nerve Sensitisation:** Deep buttock pain + positive FADIR + sciatic-type symptoms = sciatic nerve involvement within the deep gluteal space.
- **Priority 2 — Hip Internal Rotator / External Rotator Dysfunction:** FADIR positive and resisted hip abduction painful suggests piriformis/deep rotator involvement.
- **Priority 3 — Running Load Mismanagement:** Overuse aetiology — training load spike is likely the precipitant; addressing load management is essential for recovery and prevention.
- **Priority 4 — Hip Abductor Weakness:** Painful resisted hip abduction — gluteus medius weakness increases Trendelenburg gait pattern, increasing lateral pelvic drop and hip adduction/IR load on the sciatic nerve during running.
- **Priority 5 — Return to Running Goals:** Marathon is a specific, important functional goal that will drive rehabilitation programme design.

Q2: Primary Diagnosis, Two Differentials & Clinical Reasoning

Primary Diagnosis

Deep Gluteal Syndrome (DGS) / Piriformis Syndrome — sciatic nerve irritation or entrapment within the deep gluteal space (at the piriformis-sciatic nerve interface or fibrovascular bands), presenting as deep buttock pain with sciatic-type radiation, aggravated by hip flexion/adduction/internal rotation (FADIR positive) and resisted hip abduction.

Clinical Reasoning:

- Positive FADIR: Flexion, ADduction, Internal Rotation compresses the piriformis against the sciatic nerve — reproducing deep buttock pain. This is the classical provocative mechanism for deep gluteal space/piriformis pathology.
- Resisted hip abduction is painful: Piriformis acts as a hip abductor (when hip is flexed > 60°) and external rotator. Pain on resisted abduction implicates piriformis or gluteus medius as pain generators.
- Negative lumbar clearance: Effectively excludes lumbar disc herniation (L4/L5/S1) as the source of buttock/sciatic pain — confirming extra-spinal origin.
- Marathon runner: Repetitive hip flexion/extension with increasing mileage repeatedly loads the piriformis. Overuse tendinopathy/bursitis or nerve sensitisation is mechanically plausible.
- Deep quality of pain: Consistent with deep gluteal space pathology rather than superficial muscle or posterior chain involvement.

Differential 1: Proximal Hamstring Tendinopathy (PHT)

PHT produces deep buttock pain at the ischial tuberosity, aggravated by running (particularly uphill), sustained hip flexion with knee extension (stretching the hamstring origin), and sitting on hard surfaces. PHT is distinguished by: pain localising to the ischial tuberosity (palpable), pain on resisted knee flexion at 90° hip flexion, and positive bent-knee stretch test. FADIR would not typically reproduce PHT pain (it does not stress the ischial attachment). The location of pain ('deep buttock' vs. 'sitting-bone pain') and aggravating positions differentiate. In a marathon runner, PHT is extremely common and should be specifically ruled out.

Differential 2: Gluteal Tendinopathy (Greater Trochanteric Pain Syndrome)

Gluteal tendinopathy (gluteus medius/minimus at greater trochanteric insertion) produces lateral hip/buttock pain, aggravated by hip adduction positions (crossing legs, sustained hip IR, climbing stairs) and single-leg loading. Resisted hip abduction would be painful and weak. Distinguishing from DGS: location is lateral (greater trochanter) not deep central; FADIR may be painful due to compressive loading of the tendon against the trochanter; sciatic radiation is absent. The painful resisted hip abduction in this case raises this as a genuine differential. An MRI or ultrasound can distinguish between tendinopathy and DGS/nerve involvement.

Q3: Objective Assessment Techniques & Interpretation

Technique 1: FADIR Test (Flexion, Adduction, Internal Rotation)

- **Rationale:** The primary clinical test for deep gluteal space pathology and hip intra-articular pathology. Reproduces compressive loading of the piriformis/sciatic nerve interface.
- **Procedure:** Patient supine. Physiotherapist passively moves the hip into 90° flexion, then applies adduction (hip crosses midline) and internal rotation simultaneously. Maintain the combined position for 5–10 seconds. A positive test reproduces the patient's deep buttock pain.

- **Interpretation:** Reproduction of deep buttock pain = positive for DGS/piriformis syndrome. Reproduction of groin pain = positive for hip intra-articular pathology (FAI, labral tear). Pain localised to the posterior buttock only (not groin) supports extra-articular DGS. If sciatic radiation is reproduced, this further implicates neural involvement within the deep gluteal space.

Technique 2: Pace Test (Resisted Hip Abduction/External Rotation at 90° Hip Flexion)

- **Rationale:** The Pace Test specifically loads the piriformis muscle in its shortened position (hip flexed 90°, slightly abducted), isolating its function as a hip abductor and external rotator at this position.
- **Procedure:** Patient seated at edge of plinth with hips at 90° flexion. Ask patient to abduct and externally rotate both hips simultaneously against the physiotherapist's resistance applied to the lateral knees. Alternatively, perform unilaterally in side-lying.
- **Interpretation:** Pain or weakness on the affected side during resisted abduction/ER = positive Pace Test, implicating piriformis. Compare bilaterally for asymmetry. Pain without weakness is consistent with a reactive/irritable piriformis; pain with significant weakness suggests either tendinopathy or neurogenic inhibition.

Technique 3: Sciatic Nerve Palpation and Neurodynamic Assessment (SLUMP Test)

- **Rationale:** Although lumbar clearance is negative (SLR negative), the sciatic nerve may be sensitised within the deep gluteal space — a form of peripheral (extra-spinal) neural mechanosensitivity.
- **Procedure:** Palpation: With the patient prone, palpate the sciatic nerve in the deep gluteal space (midway between ischial tuberosity and greater trochanter). Apply gentle digital pressure — tenderness and/or reproduction of leg symptoms = positive. SLUMP test: patient slumped in sitting with progressive sensitisers (cervical flexion, ankle dorsiflexion) — does this reproduce posterior thigh/buttock neural symptoms beyond mechanical pain?
- **Interpretation:** Tenderness at the sciatic nerve in the deep gluteal space + neural symptom reproduction on SLUMP = peripheral neurodynamic involvement. This informs neural mobilisation (nerve gliding) as a component of treatment. Distinguishes from a purely musculotendinous problem where neurodynamic testing would be negative.

Q4: Initial & Long-Term Treatment / Management Plan

SMART Goals

- **Short-term (4–6 weeks):** Able to run 5 km pain-free; FADIR negative; pain NRS ≤ 2/10 during resisted hip abduction.
- **Long-term (16–20 weeks):** Return to full marathon training; pain-free at all hip ranges; hip abductor strength 5/5 bilaterally.

Initial Management (Weeks 1–3) — Load Modification Phase

- **Running Load Reduction:** Reduce weekly mileage by 50% immediately. Avoid hill running, speed work, and long slow distance (LSD) runs until pain settles. Replace with pain-free cross-training: swimming (non-weight-bearing hip flexion/extension), cycling (low resistance, saddle height to avoid hip adduction).
- **Modify Aggravating Postures:** Avoid prolonged sitting on hard surfaces (use cushion over ischium). Avoid hip adduction/IR in daily activities — uncross legs, adjust seat height (slightly higher seat reduces hip flexion).
- **Soft Tissue Therapy — Piriformis:** Deep tissue massage / trigger point release to piriformis and the deep external hip rotators (obturator internus, gemelli, quadratus femoris). This reduces muscle tone compressing the sciatic nerve.
- **Dry Needling:** Trigger point dry needling to piriformis is a well-supported adjunct for deep gluteal muscle hypertonicity — reduces TrP-mediated neural compression.
- **Neural Mobilisation:** Sciatic nerve gliding exercises (supine — alternating hip flexion/knee extension cycles in a pain-free range) to restore neural mechanosensitivity. Start with slider techniques before tensioner techniques.

Progressive Rehabilitation (Weeks 4–20) — Strengthening and Return to Running

- **Piriformis Stretching:** Figure-four stretch (supine), pigeon stretch (in a controlled manner), seated hip ER stretch. Hold 30–45 seconds, 3 reps, twice daily. Progress intensity gradually — avoid aggressive stretching that provokes neural symptoms.
- **Hip Abductor Strengthening:** Side-lying hip abduction, clamshells, standing band hip abduction, single-leg glute bridge, lateral band walks. Progress using FITT principle: initially 2 × 15 reps low load; progress to 3 × 10 reps moderate load; functional pattern (single-leg squat).
- **Gluteus Maximus Loading:** Bridges → RDL (Romanian deadlift) → hip thrust → kettlebell deadlift. Running requires gluteus maximus power — progressive hip extension loading is essential.
- **Running Mechanics:** Gait retraining — increase running cadence by 5–10% (reduces hip adduction at midstance), avoid contralateral pelvic drop (indicates gluteus medius weakness), avoid trunk lean toward stance leg.
- **Graded Return to Running Protocol:** Walk-run intervals → continuous easy runs (Zone 1, conversational pace) → progressive distance (10% rule: increase weekly mileage by no more than 10%) → reintroduce hills and tempo runs last.
- **Adjunct / Escalation:** Ultrasound-guided corticosteroid injection to piriformis/sciatic nerve sheath may be considered if conservative management fails at 8–12 weeks (GP/sports physician referral). MRI if symptoms not resolving to exclude structural sciatic nerve pathology (piriformis variant anatomy — a bifid piriformis can mechanically entrap the sciatic nerve and may require surgical decompression in refractory cases).

> *Key clinical message: Deep gluteal syndrome is a diagnosis of exclusion — lumbar pathology has been cleared. FADIR positive + resisted abduction pain + marathon runner context = strong clinical picture. Load management and progressive gluteal strengthening are the cornerstones of management.*

Quick Reference Summary

Case	Patient	Primary Diagnosis	Key Red Flags	Core Treatment Focus
A	Mrs. Gupta 50F, Graphic Designer	Cervicogenic Neck Pain (Upper Crossed Syndrome)	Osteoporotic fracture; CAD (5Ds 3Ns); Myelopathy	DCF strengthening; Postural correction; Ergonomic advice; Grade II–III mob (no HVLA)
B	Mr. O'Shea 72M, Retired Librarian	Osteoporotic VCF T7–T8	**URGENT:** Fracture; Malignancy; Cord compression	Refer FIRST; No manipulation; Falls prevention; Bone health optimisation
C	Ms. Taylor 32F, Yoga Instructor	Postpartum SIJ Dysfunction (PGP)	Cauda equina; DVT; Sacral stress fracture	SIJ belt; Inner unit rehab; Gluteal strengthening; Graded yoga return
D	Mr. Rossi 45M, Marathon Runner	Deep Gluteal Syndrome (Piriformis/DGS)	Femoral neck stress fracture; Pelvic malignancy	Load management; Piriformis Rx; Gluteal strengthening; Running retraining

OSCE Examination Tips

- Always screen red flags BEFORE proceeding to objective assessment — verbalise your reasoning aloud to the examiner.
- Use the ICF framework to present your problem list: Body Structure/Function → Activity → Participation.
- When presenting differentials, explain WHY each is more or less likely based on YOUR findings — not just textbook features.
- Name, demonstrate, and interpret every objective test — examiner credit requires all three elements.
- SMART goals should be specific: include time frame, measurable outcome, and functional relevance to the patient.
- Integrate safety considerations naturally: osteoporosis (no HVLA), postpartum (avoid compression tests on SIJ initially), runners (load management before strengthening).
- In your treatment plan, start with patient education and self-efficacy — consistently evidence-based best practice across all MSK conditions.

- If the examiner asks "what would change your management?" — always answer with: imaging findings, worsening neurological signs, failure to progress at 6–8 weeks.